

BRAC University
Dept. of Computer Science and Engineering
Spring 2025
CSE331-01
QUIZ - 5

Time: 25 minutes

Marks: 20

Name: _____

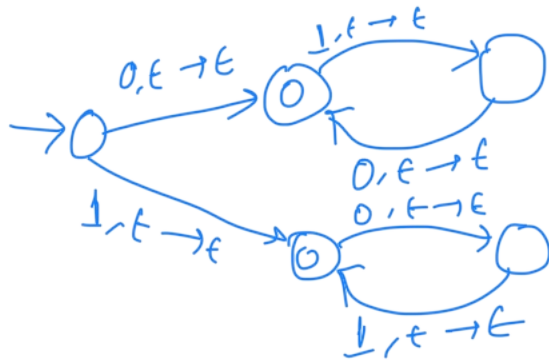
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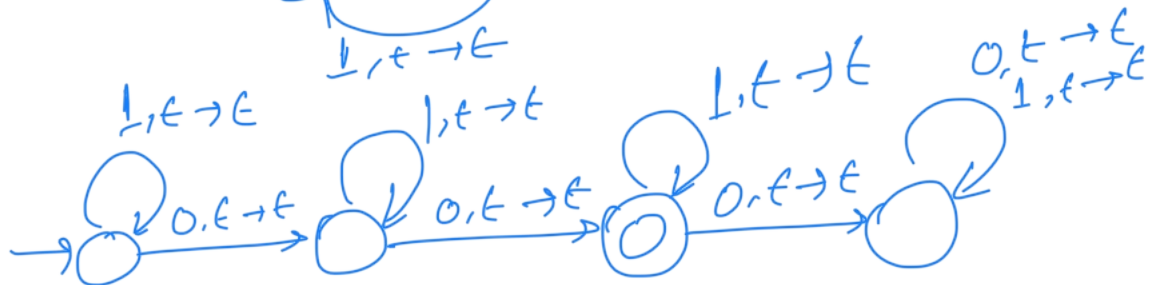
1. Construct PDA for the following languages. [7+7+6=20]

- $L = \{w \in \{0, 1\}^* : 0 \text{ and } 1 \text{ alternates in } w \text{ and length of } w \text{ is odd}\}$
- $L = \{w \in \{0, 1\}^* : \text{the count of } 0\text{'s in } w \text{ is two.}\}$
- $L = \{w \in \{a, b, c\}^* : w = c^i a^j b^k, \text{ where } i > 2k, j, k \geq 0\}$

a)



b)



c)

$$w = c^{2k} c^D a^j b^k$$

$c \quad c \quad a \quad b$

$$i = 2k + D \rightarrow 1 \text{ to } \infty$$

$$a, \epsilon \rightarrow \epsilon \quad b, x \rightarrow \epsilon$$

